Failure Analysis Report

FAIL-0226 — Synchronization Failure

1. Failure Summary

Failure ID	FAIL-0226
Failure Type	Synchronization Failure
Severity	Minor
Component	Health API
Application	APP-002 — MobileCommunication
Build Version	6.3.8
Timestamp	2024-01-13 00:51:40
Environment	ENV-008 — Production-like
OS Version	One UI 6.0
Network	5G
Resolution	Wont Fix

2. Test Case Reference

Test Case ID	TC-1016
Title	Validate that configuration reload returns the correct result when the battery l...
Execution ID	EXEC-01062
Test Run	RUN-006 — Regression Cycle 6 — Build 5.2.1
Requirement	REQ-0064 — Compatibility — Remote Diagnostics

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete push notification delivery within the defined SLA.
Observed Behaviour	The system exhibited synchronization failure when the input data contains Unicode special characters...

4. Error Details

Error Message:

```
SyncError: conflict detected on entity 'ENTITY-5327'; last_modified timestamps diverge by 21s.
```

Stack Trace / Log Excerpt:

```
com.samsung.wallet.UILayer.execute(line 177)
```

Log Reference	LOG-36467

5. Executed Steps at Time of Failure

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1	Capture a screenshot and attach as evidence.	A success toast message is displayed.
2	Re-connect the headset and confirm auto-recovery.	No crash or ANR is observed.
3	Disconnect the wearable and wait 20 seconds.	The operation completes without error.
4	Record response time using the built-in performance monitor.	A confirmation notification appears on the paired device.

6. Root Cause Analysis

Health sensor sampling rate exceeds Bluetooth LE characteristic notification limit of 250ms intervals.

8. Historical Similar Failures

HIST-0225	FAIL-0226	Synchronization Failure	6.3.9	2021-09-11	2	Resolved
HIST-0288	FAIL-0289	Synchronization Failure	3.5.9	2021-03-03	12	Wont Fix
HIST-0380	FAIL-0381	Synchronization Failure	6.5.9	2021-07-31	11	Open

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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